Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
7	(a)			Si (1) award (1) for answer that identifies one property as metallic and another as non-metallic e.g. it has a high melting point but is brittle neutral answers it is a semiconductor it has metal and non-metal properties		2		2		
	(b)			The density of metals and non-metals increases ☐ The boiling point of metals increases but the boiling point of non-metals shows no trend ✓ The density of metals shows no trend but the density of non-metals decreases ☐ The boiling point of metals and non-metals shows no trend ☐ The density of metals increases but the density of non-metals shows no trend ✓ The boiling point of metals shows no trend but the boiling point of non-metals decreases ☐ The density of metals decreases but the density of non-metals shows no trend ☐			2	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			there is no trend in the melting points of the non-metals / the elements preceding chlorine accept description e.g. there is a decrease in melting point from Si to P, then an increase from P to S and then another decrease from S to Cl neutral answer – melting point is unpredictable			1	1		
	(d)	(i)		liquid (1) award (1) for any of following only if first mark is awarded • 60 °C is <u>between</u> its melting point and boiling point • melting point is below 60 °C and boiling point is above 60 °C • 60 °C is <u>between</u> 44 °C and 281 °C • phosphorus has already melted at 60 °C but has not reached its boiling point neutral answer – its melting point is 44 °C and its boiling point is 281 °C			2	2		
		(ii)		$\boxed{3}\,Zn + \boxed{2}\,H_3PO_4 \longrightarrow Zn_3(PO_4)_2 + \boxed{3}\,H_2$		1		1	1	
				Question 7 total	**0**	**3**	**5**	**8**	**1**	**0**